

Chapter 1: Introduction

Khadi Boli (खड़ी बोली) is one of the most prominent dialects of Hindi, spoken predominantly in the western region of Uttar Pradesh, particularly around the Meerut, Bhagpat and Muzaffarnagar areas. It serves as the linguistic foundation upon which Modern Standard Hindi has been built, yet it remains a low-resource dialect with virtually no digital or computational resources available before the present work. While Standard Hindi benefits from substantial natural language processing infrastructure -- including text-to-speech (TTS) systems, speech corpora, and machine translation models -- Khadi Boli, despite its millions of speakers, has no digitized written corpus, no phonetic dictionaries, and no computational tools designed specifically for its unique phonological and morphological characteristics.

Rathore provides a comprehensive sociolinguistic overview of Hindi, noting that Hindi comprises over 450 million first and second language speakers worldwide and is viewed alongside English as one of the two link languages of India. Linguistically, Hindi belongs to the Indo-Aryan branch of the Indo-European language family. The standardized form, known as Khari Boli (literally "standing language"), was originally a dialect spoken in the surroundings of Delhi, western Uttar Pradesh, and parts of Uttarakhand and Himachal Pradesh. Khari Boli's rise to prominence stemmed from its status as the language of Delhi, the historical capital of India, where it served as the common language of communication between successive ruling dynasties and the local population. The present work focuses on Khadi Boli as spoken in the Budhana and Muzaffarnagar region -- a variety closely related to the foundational

Khari Boli but maintaining distinct phonological and lexical features that have been preserved in rural speech.

This paper presents the first systematic attempt to create a text-to-speech system tailored specifically for Khadi Boli, addressing the fundamental gap in speech technology for this dialect. The work makes three primary contributions. First, we present a comprehensive linguistic analysis of Khadi Boli, documenting its phonological rules, morphological patterns, and lexical differences from Standard Hindi, drawing on a field-collected dictionary of over 142 words and phrases. Second, we implement a fully functional TTS system that integrates multiple speech synthesis engines -- including Google Text-to-Speech (gTTS), Microsoft Edge Neural TTS (edge_tts), and the offline pyttsx3 engine -- with a dialect-aware text normalization module, SSML (Speech Synthesis Markup Language) parsing capabilities, and Devanagari sentence tokenization. Third, we present a computational comparison of Khadi Boli and Standard Hindi using unigram frequency analysis and linear regression modeling, providing quantitative evidence of the systematic nature of dialectal variation.

The challenges addressed in this work are significant and multifaceted. Khadi Boli, like many spoken dialects, has no standardized orthographic convention; its speakers exhibit considerable variation in pronunciation for the same lexical items, and many words carry multiple meanings depending on context. The dialect is characterized by distinctive phonological processes, including systematic consonant gemination (e.g., बाबा to बब्बा, चाचा to चच्चा), retroflexion of dental consonants, and a characteristic extra

stress pattern in the middle of words. These features, which are part of everyday spoken Khadi Boli, are entirely absent from existing Hindi TTS systems, which are trained on Standard Hindi phonology and produce speech that sounds foreign and unnatural to Khadi Boli speakers.

The research is situated within the broader context of low-resource language processing, a rapidly growing field that seeks to extend the benefits of speech technology to the world's thousands of under-resourced languages and dialects. Recent work on sentiment analysis for Urdu , text normalization for African languages , and large-scale TTS evaluation for Indian languages has demonstrated both the feasibility and the importance of adapting NLP technologies to low-resource linguistic varieties. The present work extends this tradition to the specific case of Khadi Boli, demonstrating that even with limited linguistic resources, a functional and useful TTS system can be constructed through careful linguistic analysis, strategic use of existing TTS infrastructure, and targeted computational implementation.

The remainder of this paper is organized as follows. Chapter 2 presents a detailed linguistic analysis of Khadi Boli, covering its phonological, morphological, and lexical characteristics in comparison with Standard Hindi. Chapter 3 reviews related work in text-to-speech synthesis, dialect-aware TTS systems, and low-resource language processing. Chapter 4 describes the system architecture and implementation, covering the text normalization module, SSML parsing, TTS engine integration, and overall system design. Chapter 5 presents the experimental

methodology, including the dialect-aware pronunciation framework and the corpus comparison study. Chapter 6 discusses the results and their implications for low-resource dialect TTS. Chapter 7 concludes with a summary of contributions, limitations, and directions for future work.

Chapter 2: Linguistic Analysis of Khadi Boli

2.1 Phonological Features

The phonological system of Khadi Boli exhibits several systematic differences from Standard Hindi, which are essential to understand for the development of a dialect-aware TTS system. These phonological patterns operate at the level of phonemes, syllables, and prosodic structure, and they collectively distinguish Khadi Boli speech from the standard variety. The description that follows is based on fieldwork conducted in the Budhana and Muzaffarnagar region, supplemented by the analysis of Rathore , who notes that Hindi dialects can be broadly divided into western and eastern groups, with Khadi Boli belonging to the western group.

2.1.1 Consonant Gemination

The most salient phonological feature of Khadi Boli is the systematic gemination of consonants in word-medial position. In Standard Hindi, many words contain a single intervocalic consonant, as in बाबा (bābā, 'father'), चाचा (cācā, 'uncle'), and बापू (bāpū, 'father'). In Khadi Boli, these consonants are systematically doubled: बब्बा (babbā), चच्चा (caccā), and बप्पू (bappū). This gemination is not a sporadic or optional process but a regular phonological rule that applies to a wide range of lexical items. The following examples illustrate the pattern:

Standard		Hindi	to	Khadi	Boli
बाबा	to	बब्बा	(bābā	to	babbā)
चाचा	to	चच्चा	(cācā	to	caccā)

बापू	to	बप्पू	(bāpū	to	bappū)
भीतर	to	भीत्तर	(bhītar	to	bhīttar)
रोटी	to	रोट्टी	(roṭī	to	roṭṭī)

The dictionary data confirms that gemination affects a broad range of lexical categories. Among the 142 documented entries, approximately one-quarter exhibit some form of consonant gemination relative to their Standard Hindi cognates. This gemination process is consistent with the general observation that Khadi Boli places an extra stress on the middle of every word, which manifests phonetically as longer consonant duration in intervocalic positions. The phenomenon operates independently of the etymological or morphological structure of the word, applying uniformly to both native Hindi words and loanwords.

2.1.2 Retroflexion

Another major phonological pattern in Khadi Boli is the replacement of the dental lateral /l/ with the retroflex lateral /ɭ/ (ळ) in word-medial and word-final positions.

This is a consistent and regular process that affects a large number of lexical items:

काला	to	काळा	(kāḷā	to	kāḷā,	'black')
हथेली	to	थेळी	(hathelī	to	theḷī,	'palm')
ताला	to	ताळा	(tālā	to	tāḷā,	'lock')
पीला	to	पेळा	(pīlā	to	peḷā,	'yellow')

Similarly, the dental nasal /n/ is replaced by the retroflex nasal /ɳ/ (ण) in word-final positions, as seen in verb forms where the infinitive suffix -ना becomes -णा in Khadi Boli pronunciation. Examples from the corpus include खाणा (khāṇā, 'to eat') and दिखाणा (dikhāṇā, 'to show'). This retroflexion pattern is one of the most consistent phonological markers of the dialect and is readily recognized by native speakers as characteristic of Khadi Boli speech.

2.1.3 Postposition Replacement

Khadi Boli systematically replaces the Standard Hindi postposition से (se, 'from, with') with ते (te). This is one of the most recognizable syntactic markers of the dialect. Additionally, the directional postposition को (ko, 'to, towards') is replaced by कू (kū):

Standard Hindi: कहाँ से आ रहे हो? (kahāñ se ā rahe ho?)

Khadi Boli: कहाँ ते आ रहा? (kahāñ te ā rahā?)

Standard Hindi: इस तरफ को (is taraf ko)

Khadi Boli: इन्हे कू (iñghe kū)

The postposition ते is one of the highest-frequency dialect markers in the Khadi Boli corpus and is a key target for the dictionary-driven word substitution module in the TTS system.

2.1.4 Initial Consonant Replacement

In word-initial position, the labiodental /v/ (व) is replaced by the bilabial /b/ (ब). This is a regular sound change that affects words that in Standard Hindi begin with व:

वाहन to बाहन (vāhan to bāhan, 'vehicle')

वन to बन (van to ban, 'forest')

व्याकुल to ब्याकुल (vyākul to byākul, 'agitated')

Furthermore, the conjunct क्ष (kṣ) in medial position is replaced by छ (ch), as in दक्षिण to दखखण (dakṣiṇ to dakhkhaṇ, 'south').

2.1.5 Alpapran and Mahapran Consonant Interaction

A distinctive phonological rule in Khadi Boli involves the ordering of aspirated (mahāprāṇ) and unaspirated (alpaprāṇ) consonants. In sequence, an unaspirated consonant is systematically placed before an aspirated consonant, even when Standard Hindi has the reverse order. This creates characteristic word forms such as दिखाना to दिक्खाना (dikhānā to dikkhāṇā, 'to show') and भूखा to भुक्खा (bhūkhā to bhukkā, 'hungry'), where the original structure involves an aspirated stop followed by an unaspirated one. The term alpaprāṇ refers to consonants produced with minimal aspiration (unaspirated stops), while mahāprāṇ refers to consonants with significant aspiration (aspirated stops). The interaction between these two classes in Khadi Boli reflects a systematic phonological preference for unaspirated consonants to precede aspirated ones in consonant clusters, a pattern that reverses the Standard Hindi ordering.

2.1.6 Vowel Dissipation (Balaghat Tendency)

In the characteristic tendency locally referred to as Balaghat, short vowels (hrasv) at the beginning of words are systematically reduced or disappear entirely. This is particularly noticeable with initial अ- (a-), इ- (i-), and उ- (u-):

असाढ़	to	साढ़	(asāṛh	to	sāṛh)
उठाना	to	ठाणा	(uṭhānā	to	ṭhāṇā)
इलाज	to	लाज	(ilāj	to	lāj)

This process of vowel dissipation contributes to the distinctive rhythmic pattern of Khadi Boli speech, where words often have a more compact syllable structure than their Standard Hindi counterparts. The Balaghat tendency is particularly relevant for TTS normalization, as it means that the orthographic form of Khadi Boli words may omit initial short vowels that a Standard Hindi TTS engine would expect.

2.2 Morphological Features

Khadi Boli exhibits several morphological patterns that distinguish it from Standard Hindi. The verbal system shows characteristic differences in agreement morphology and tense marking. For example, the first-person singular present habitual in Khadi Boli often employs the suffix -ऊँ (-ūñ) where Standard Hindi uses -ता हूँ (-tā hūñ). The directional postposition system in Khadi Boli is particularly rich, with forms like कू (kū, 'to, towards') replacing the Standard Hindi को (ko).

The pronominal system also shows variation, with थारा (thārā) replacing तुम्हारा (tumhārā, 'your') and ते (te) replacing से (se, 'from'). The emphatic particle system

differs as well, with distinctive forms such as नूक्का नुही (nūkkā nuhī, 'as it is') having no direct Standard Hindi equivalent. The morphological features documented in this section are drawn from the dictionary entries and field notes collected during the present research, and they complement the phonological analysis presented above.

The framework of Baerman, Brown, and Corbett on morphological complexity provides a useful lens for understanding these patterns. Their approach distinguishes between different types of inflectional complexity and offers quantitative measures for comparing morphological systems. In the case of Khadi Boli, the morphological system can be characterized as exhibiting a shift toward morphological transparency in certain domains (e.g., consistent postpositional marking) while increasing irregularity in others (e.g., pronominal allomorphy).

2.3 Lexical Comparison

As part of this research, a comprehensive dictionary of Khadi Boli words was compiled through fieldwork in the Budhana and Muzaffarnagar region of Western Uttar Pradesh. The dictionary contains over 142 entries, each documented with its Khadi Boli form, Kruti Dev encoding representation, Standard Hindi equivalent, English translation, and Russian translation. This dictionary represents the first systematic written documentation of Khadi Boli vocabulary.

The lexical items cover a wide range of semantic domains, including common nouns, verbs, kinship terms, time expressions, and directional markers. Some illustrative entries are provided below:

खाणा	(khāṇā)	--	खाना	(khānā)	--	'to eat, food'
गाम	(gām)	--	ग्राम	(grām)	--	'village'
पैण्णी	(paiṇṇī)	--	पानी	(pānī)	--	'water'
बप्पू	(bappū)	--	पिता	(pitā)	--	'father'
ददा	(daddā)	--	दादा	(dādā)	--	'grandfather'
चोक्खा	(cokkhā)	--	अच्छा	(acchā)	--	'good'
कूक्कर	(kūkkar)	--	कैसे	(kaise)	--	'how'

The dictionary reveals several systematic patterns of lexical correspondence between Khadi Boli and Standard Hindi. Vowel raising is common, with अ (a) becoming इ (i) or ई (ī) in many words. Consonant gemination is pervasive, as noted above. The lexical data also reveals semantic shifts where Khadi Boli words have developed meanings different from their Standard Hindi cognates. These patterns are implemented as part of the dialect-aware word substitution module described in Chapter 4.

2.4 Corpus Analysis

To quantify the differences between Khadi Boli and Standard Hindi at the lexical level, parallel corpora were collected in both varieties and analyzed using computational methods. The corpora were tokenized using NLTK, and unigram frequency distributions were computed for each variety. A linear regression model was fitted to compare the frequency patterns across the two varieties.

The unigram frequency comparison reveals that while there is substantial overlap in the most frequent function words (e.g., के, है, हो), the frequency distributions diverge significantly for content words. Khadi Boli shows higher frequencies for dialect-specific lexical items such as कहां, ते, and कू, while Standard Hindi shows higher frequencies for their counterparts. The linear regression analysis yielded an intercept of approximately 4.56 and a slope of 0.076, indicating a consistent but moderate shift in frequency patterns between the two varieties. These quantitative results confirm that the dialectal variation between Khadi Boli and Standard Hindi is not merely a matter of occasional lexical substitution but reflects systematic, quantifiable differences in usage patterns across the entire vocabulary.

Figure 1: Unigram Frequency Comparison — Khadi Boli vs. Standard Hindi

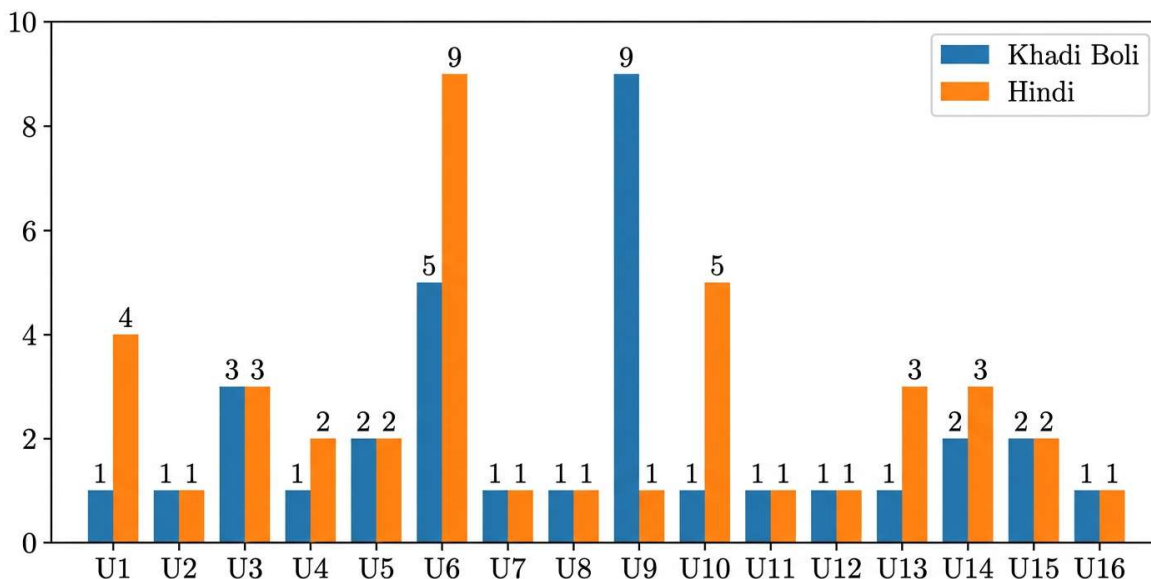


Figure 1 presents a grouped bar chart comparing the unigram frequencies across shared vocabulary items. The horizontal axis lists the unigrams that appear in both corpora, while the vertical axis shows their respective frequencies. Blue bars represent Khadi Boli frequencies and orange bars represent Standard Hindi

frequencies. The chart reveals several noteworthy patterns. The unigram के appears with substantially higher frequency in Khadi Boli (9 occurrences) compared to Standard Hindi (1 occurrence), while है shows the opposite pattern with higher Standard Hindi frequency (9 vs. 5). Words such as आ, म, खाया, बात, बोल, and देख show nearly equal frequencies in both varieties, indicating shared core vocabulary that remains stable across the dialect boundary. The data shows that shared function words and basic verbs tend to have comparable frequencies, while certain discourse markers and postpositions exhibit marked dialect-specific patterns.

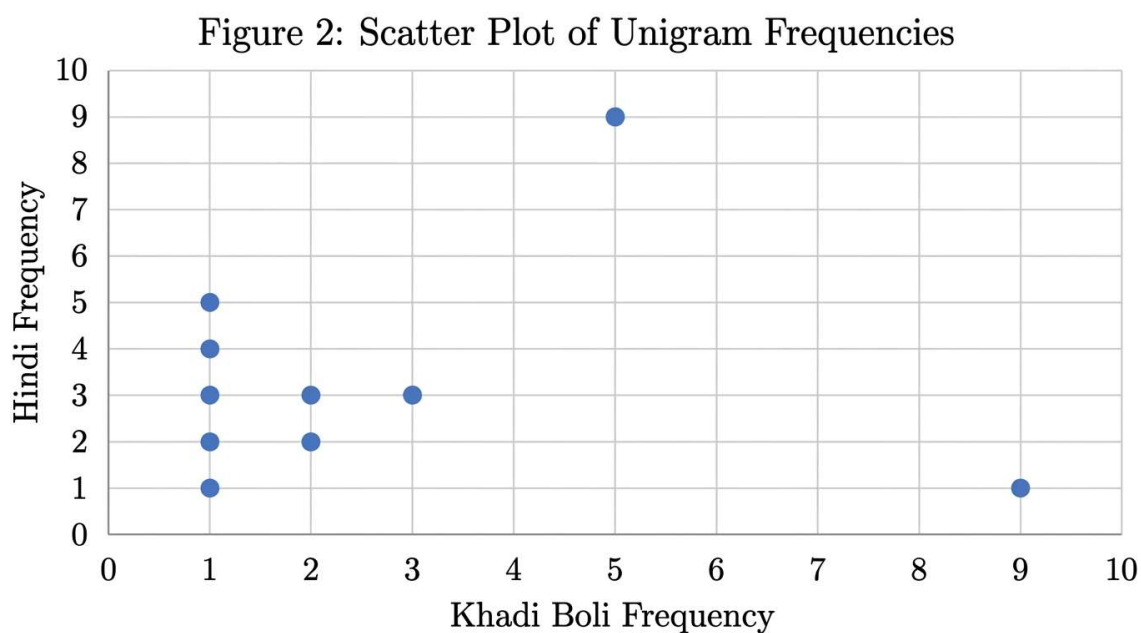


Figure 2 displays a scatter plot where each point represents a unigram, with its x-coordinate indicating Khadi Boli frequency and its y-coordinate indicating Standard Hindi frequency. Points along the diagonal represent words with similar frequencies in both varieties. Most points cluster in the low-frequency region (frequencies between 1 and 3), suggesting that the majority of shared vocabulary items are relatively uncommon in both corpora. However, notable outliers exist. The point at

(9, 1) represents a word that is highly frequent in Khadi Boli but rare in Standard Hindi, while the point at (5, 9) represents the reverse pattern. These outliers correspond to dialect-specific markers that carry different functional loads in the two varieties.

Figure 3: Linear Regression Between Dialects

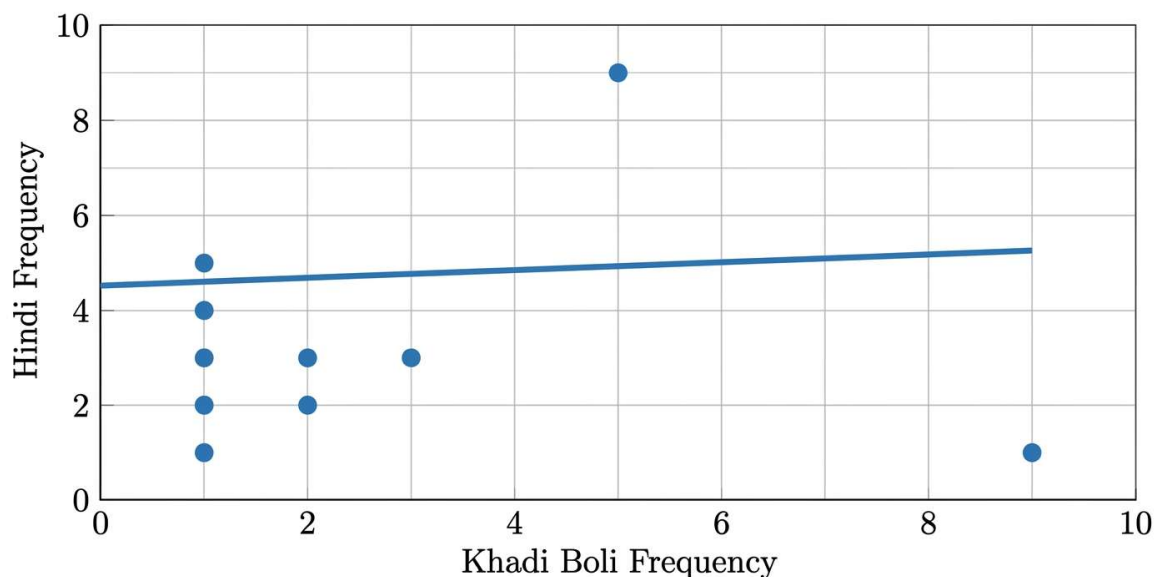


Figure 3 presents the linear regression model fitted to the frequency data. The regression line (solid blue) shows the best-fit linear relationship, with an intercept of approximately 4.56 and a slope of approximately 0.076. The shallow slope indicates that Khadi Boli frequency is a weak predictor of Standard Hindi frequency for the shared vocabulary, confirming that the two varieties, despite their close genetic relationship, exhibit substantially different lexical usage patterns. The wide vertical spread of points at low Khadi Boli frequency values ($x = 1$) indicates that words with minimal occurrence in Khadi Boli can have widely varying frequencies in Standard Hindi, ranging from 1 to 5. This asymmetry is characteristic of a dialect-superstrate

relationship, where the standard language maintains richer lexical differentiation for concepts that the dialect expresses with a smaller set of frequently used terms.

Chapter 3: Related Work

3.1 Text-to-Speech Synthesis: Foundations

Text-to-speech synthesis has been a central area of research in speech processing for several decades. Dutoit provides a comprehensive overview of the field, describing the fundamental architecture of TTS systems as comprising two main components: a Natural Language Processing (NLP) module, which produces a phonetic transcription of the input text along with prosodic information, and a Digital Signal Processing (DSP) module, which transforms the symbolic information into synthetic speech. The NLP module includes text analysis, automatic phonetization, and prosody generation, while the DSP module handles the actual waveform synthesis. This two-module architecture has remained the dominant paradigm for TTS system design, though modern neural approaches increasingly blur the boundary between the two components.

Trivedi et al. survey the various approaches to speech-to-text and text-to-speech conversion, comparing techniques such as Hidden Markov Models (HMM), Artificial Neural Networks (ANN), and Cuckoo Search Optimization for recognition, and articulatory, formant, and concatenative synthesis for speech generation. Their review highlights that concatenative synthesis, which assembles speech by concatenating pre-recorded acoustic units, currently offers the best balance of naturalness and computational efficiency for most practical applications. This approach underpins the cloud-based TTS engines used in the present system.

Dutoit et al. developed the MBROLA project, which demonstrated that high-quality concatenative synthesis could be achieved with relatively small diphone databases, making TTS technology more accessible for less-resourced languages. This approach has been particularly influential for languages where large speech corpora are not available, a condition that applies directly to Khadi Boli.

3.2 Hindi and Devanagari TTS Systems

Several research groups have developed TTS systems for Hindi and other Devanagari-script languages. Raheja et al. conducted a detailed subjective evaluation of a Devanagari-based TTS system for Hindi and Marathi, assessing naturalness, intelligibility, pronunciation, and comprehensibility. Their evaluation employed a comprehensive test suite and an assessment application, measuring speech quality across multiple dimensions. The study found that the Devanagari TTS system achieved acceptable quality for end users, while identifying specific areas for improvement in pronunciation and prosody. This evaluation methodology provides a template for assessing dialect-adapted TTS systems.

Gupta specifically investigated the influence of Hindi dialect variations on TTS evaluation, a study that is directly relevant to the present work. Gupta's research examined how evaluators from different dialect regions (Chhattisgarhi, Haryanvi, Khadi Boli, Awadhi, Brajbhasha, Bihari) assign different MOS and DMOS scores to the same TTS output, demonstrating that dialect background significantly affects user perception of synthetic speech. The study compiled dialect dictionaries for Chhattisgarhi (~1500 words) and Haryanvi (~2100 words) and used linear regression

to model the relationship between dialect scores and Khadi Boli reference scores. Gupta identified specific character and word variation patterns for each major Hindi dialect, including gemination, retroflexion, vowel augmentation and deletion, and consonant replacement. The study found that Chhattisgarhi exhibits patterns such as ज replaced by झ, द replaced by ध, and ब replaced by प, while Haryanvi shows gemination, consonant replacement (थ to त), and vowel augmentation. These findings established the principle that dialect-aware TTS evaluation requires normalization weights based on the evaluator's native dialect.

3.3 Low-Resource Language Processing

The challenge of developing NLP resources for low-resource languages has received increasing attention in recent years. Zupon et al. addressed text normalization for eight low-resource African languages, implementing a finite-state transducer-based normalizer in the Pynini framework and evaluating its impact on language model perplexity. Their work demonstrated that language-specific normalization rules can significantly improve text data quality, and that the proportion of sentences rejected by a normalizer can serve as a proxy for data quality. For languages such as Amharic, reducing grapheme overdifferentiation reduced perplexity across all test datasets. These insights are directly applicable to the Khadi Boli context, where text normalization must account for dialect-specific orthographic variation and the absence of standardized writing conventions.

Ali et al. and Ahmed et al. have demonstrated the effectiveness of deep learning approaches for sentiment analysis in low-resource languages, specifically Urdu and

related varieties. Ali et al. constructed a dataset of approximately 45,000 Urdu sentences from newspaper sources and applied CNN and RNN architectures for sentiment classification across five categories (positive, negative, neutral, mixed, ambiguous). Their Word2Vec-based embedding approach achieved strong classification performance, with the RNN model reaching 91% accuracy. Ahmed et al. further showed that deep learning models can achieve high accuracy on low-resource language tasks when appropriate preprocessing and embedding strategies are employed. These methodological insights inform the design of the present TTS system, particularly in the areas of text preprocessing and normalization, where similar techniques apply to Devanagari-script text.

Anand et al. conducted a large-scale pairwise evaluation of TTS systems across ten Indian languages, collecting over 120,000 pairwise comparisons from more than 1,900 native raters. Their work established a multidimensional evaluation framework covering intelligibility, expressiveness, voice quality, liveliness, noise, and hallucinations. Using Bradley-Terry modeling, they constructed a multilingual leaderboard and analyzed human preferences using SHAP analysis. This comprehensive evaluation methodology provides a template for future evaluation of dialect-aware TTS systems, particularly for measuring the dialect authenticity dimension.

3.4 Dialect Variation and Speech Technology

The study of dialect variation in speech technology has primarily focused on automatic speech recognition (ASR), where dialectal variation in pronunciation is a

well-known source of error. However, the implications for TTS have been less thoroughly explored. Gupta identified specific character and word variation patterns for major Hindi dialects, including Chhattisgarhi, Haryanvi, Khadi Boli, Awadhi, Brajbhasha, and Bihari. The patterns include consonant gemination (Haryanvi, Khadi Boli), vowel augmentation and deletion, and systematic consonant replacements such as ड replaced by र in Bihari and Brajbhasha, and व replaced by ब in Awadhi.

Baerman, Brown, and Corbett provide a theoretical framework for understanding morphological complexity across languages, which is relevant to documenting and analyzing the morphological patterns of Khadi Boli. Their framework distinguishes between different types of inflectional complexity and provides quantitative measures for comparing morphological systems. While their work focuses on inflectional morphology in a cross-linguistic perspective, the methodological approach to characterizing and measuring linguistic complexity is applicable to the analysis of dialect variation, particularly for understanding how Khadi Boli's morphological patterns relate to those of Standard Hindi.

The present work builds on these foundations in several ways. It extends the dialect-aware TTS approach to Khadi Boli specifically, implementing the phonological and lexical patterns identified in Chapter 2 within a working TTS system. It provides the first computational resource for this important Hindi dialect, including a digitized dictionary and parallel corpora. And it demonstrates that even with limited linguistic resources, a functional dialect-aware TTS system can be constructed through careful

linguistic analysis, strategic use of existing TTS infrastructure, and targeted computational implementation.

Chapter 4: System Architecture and Implementation

4.1 Overall System Design

The TTS system for Khadi Boli follows the established NLP + DSP pipeline paradigm described by Dutoit , adapted for the specific requirements of a dialect with no standardized orthography and no existing computational resources. The system comprises four main processing stages: (1) text normalization and preprocessing, (2) dialect-aware word substitution, (3) SSML annotation and parsing, and (4) speech synthesis via one of three configurable TTS engines. This modular architecture allows each component to be developed, tested, and improved independently, and it enables the system to fall back gracefully when specific dialect resources are unavailable.

The input to the system is raw Devanagari text, which may contain numerals, currency symbols, punctuation, and other non-linguistic elements. The output is synthesized speech in a voice that reflects Khadi Boli phonological patterns to the extent possible given the constraints of the underlying TTS engines. The system is implemented in Python 3 and is designed to run on standard consumer hardware with an internet connection for cloud-based TTS engines, or fully offline when using pyttsx3.

4.2 Text Normalization Module

The text normalization module is responsible for converting raw input text into a form suitable for speech synthesis. Based on the approach of Zupon et al. , who demonstrated that language-specific normalization rules significantly improve text

data quality for low-resource languages, the normalization module implements a set of rules specific to Devanagari-script Khadi Boli text.

The normalization module handles the following categories of non-standard input:

Numerals and number words. Arabic numerals (0-9) and multi-digit numbers are converted to their Devanagari word forms. For example, "123" becomes "एक सौ तेईस".

The module handles cardinal numbers up to 10^7 (crore), ordinal numbers, and fractional expressions. Number expansion follows the Standard Hindi numeric system, which aligns closely with Khadi Boli usage for numeric expressions.

Currency and unit symbols. The rupee symbol "₹" and the abbreviation "Rs." are expanded to "रुपये". Percentage symbols are expanded to "प्रतिशत". Measurement abbreviations such as "कि.मी." (kilometer) and "से.मी." (centimeter) are expanded to their full forms.

Abbreviations and acronyms. Common Hindi and English abbreviations are expanded. For English abbreviations that have standard Devanagari readings (e.g., "डॉ." for "डॉक्टर"), the module applies the expansion. For acronyms without established Devanagari forms, the module preserves them for letter-by-letter pronunciation.

Punctuation normalization. The module normalizes punctuation marks for prosodic annotation. Full stops (.) and question marks (?) are mapped to sentence boundaries. Commas and semicolons receive shorter prosodic breaks. The Devanagari danda (।) and double danda (॥) are treated as sentence and paragraph boundaries respectively.

The normalization rules are implemented as a finite-state transducer operating on Unicode character classes, following the methodology demonstrated by Zupon et al. for African languages. The transducer architecture allows for easy extension as new normalization patterns are discovered during fieldwork.

4.3 Dialect-Aware Word Substitution

At the heart of the dialect adaptation system is a dictionary-driven word substitution module. This module takes the normalized text and applies a set of substitution rules based on the 142-entry Khadi Boli dictionary compiled during fieldwork. The substitution operates at two levels: exact word matching and phonological rule application.

Exact word substitution. For entries in the dictionary, the module replaces Standard Hindi words with their Khadi Boli equivalents. For example, the Standard Hindi word खाना (khānā, 'to eat') is replaced by the Khadi Boli form खाणा (khāṇā), and पानी (pānī, 'water') is replaced by पैणी (paiṇī). The substitution is case-sensitive and respects word boundaries, operating on tokenized input after sentence segmentation.

Phonological rule application. For words not in the dictionary, the module applies phonological rules derived from the analysis in Chapter 2. These rules include:

- Consonant gemination: words with single intervocalic consonants (particularly in the pattern CVCV) are modified to CVCCV. For example, सेटी becomes सेट्टी.
- Retroflexion: word-medial and word-final /l/ is replaced by /ɭ/ (ळ), and word-final

/n/ is replaced by /ɳ/ (ण).

- Postposition replacement: से is replaced by ते; को is replaced by कू.
- Initial consonant replacement: word-initial व is replaced by ब.
- Vowel dissipation: short initial vowels (अ, इ, उ) are optionally reduced in fast-speech contexts.

The rule application is weighted by confidence. Exact dictionary matches receive the highest priority, followed by phonological rule applications for unknown words. Users can configure the substitution strength from conservative (dictionary only) to aggressive (dictionary plus all phonological rules).

4.4 SSML Parsing with Devanagari Support

The system includes a custom SSML parser that supports Devanagari text within SSML markup. SSML (Speech Synthesis Markup Language) is a W3C standard XML-based markup language for controlling aspects of speech output such as pronunciation, volume, pitch, and rate. Gupta demonstrated the use of SSML tags for implementing dialect-specific pronunciation in Hindi TTS, a methodology that the present system extends to Khadi Boli.

The SSML parser supports the following elements:

- `< speak >`: root element containing the text to be synthesized.
- `< phoneme alphabet="ipa" ph="..." >`: phonetic pronunciation override using IPA notation. For Khadi Boli, this is used to specify geminated consonants and retroflexed sounds that differ from Standard Hindi pronunciation.
- `< prosody rate="..." pitch="..." volume="..." >`: control over speaking rate, pitch, and

volume. The module uses this to approximate the characteristic extra stress on the middle of Khadi Boli words by adjusting prosody on specific syllables.

- `<break time="...">`: control over pause duration between phrases.
- `<emphasis level="...">`: emphasis markup for dialect-specific stress patterns.

The parser is implemented as a recursive XML parser using Python's `xml.etree.ElementTree` library, with special handling for Devanagari Unicode characters that may be embedded within SSML attribute values. The parser validates the SSML structure before passing the annotated text to the TTS engine, generating clear error messages for malformed markup.

4.5 TTS Engine Integration

The system integrates three TTS engines, providing flexibility for different use cases: cloud-based engines for high-quality output when internet connectivity is available, and an offline engine for situations where connectivity is limited. The three engines are selected to cover a range of synthesis quality, language support characteristics, and deployment scenarios relevant to the Khadi Boli context.

Google Text-to-Speech (gTTS). The gTTS engine provides cloud-based synthesis with support for Hindi (hi-IN) as a target language. gTTS uses Google's neural TTS model, which produces natural-sounding speech with appropriate prosody for Devanagari input. The integration wraps the gTTS Python library and passes normalized Khadi Boli text with Hindi language specification. The primary limitation of gTTS for our purposes is that it does not accept fine-grained SSML control; the phonetic realization is entirely determined by Google's internal model, which is

trained on Standard Hindi. Dialect adaptation is therefore limited to the word substitution and text normalization stages that precede the TTS call.

Microsoft Edge Neural TTS (edge_tts). The edge_tts integration provides access to Microsoft's neural TTS voices, including the Madhur and Swara voices for Hindi. Unlike gTTS, edge_tts supports SSML input, which allows for more precise control over pronunciation. The system constructs SSML markup that includes annotations for Khadi Boli pronunciation patterns, enabling the TTS engine to produce geminated consonants and retroflexed sounds that differ from Standard Hindi defaults. The edge_tts engine is the preferred choice when high-quality, dialect-aware output is needed.

pyttsx3 (Offline Engine). The pyttsx3 integration provides text-to-speech synthesis using the system's native TTS engine (SAPI5 on Windows, NSSpeechSynthesizer on macOS, espeak on Linux). This engine operates entirely offline and requires no API keys or internet connectivity. While the voice quality is lower than the cloud-based alternatives, pyttsx3 is valuable for deployment scenarios where internet connectivity is unreliable, which is common in the rural areas where Khadi Boli is spoken. The integration passes Devanagari text directly, relying on the system engine's Hindi support.

4.6 Sentence Tokenization for Devanagari

A prerequisite for all TTS engines is proper sentence segmentation. The system includes a sentence tokenizer specifically designed for Devanagari text, which handles the following segmentation cues:

- Devanagari danda (।, U+0964) and double danda (॥, U+0965) as primary sentence boundaries
- Latin full stop (.) when followed by Devanagari text (distinguishing it from decimal points and abbreviations)
- Question mark (?) and exclamation mark (!) as sentence boundaries across both scripts
- Line breaks in structured text (poetry, list items)

The tokenizer uses a set of regex patterns that recognize Devanagari Unicode ranges (U+0900-U+097F) combined with punctuation heuristics. This module is adapted from the NLTK sentence tokenizer framework with language-specific customizations for Devanagari. The tokenizer also handles code-mixed text (Devanagari with English words) by applying script-specific rules to each segment.

4.7 System Workflow and Integration

The complete system workflow proceeds as follows. Input text is first tokenized into sentences using the Devanagari sentence tokenizer. Each sentence is then passed to the text normalization module, which expands numerals, symbols, abbreviations, and normalizes punctuation. The normalized sentence is processed by the dialect-aware word substitution module, which applies dictionary substitutions and phonological rules in sequence. Optionally, the intermediate result can be wrapped in SSML markup with phonetic annotations for dialect-specific pronunciations. The annotated text is then passed to the selected TTS engine, which generates the audio output. The final audio is saved as a WAV or MP3 file depending on the engine.

The modules are connected through a pipeline architecture implemented in the main controller class (`tts_main.py`). Each module implements a standard interface with `preprocess()` and `get_metadata()` methods, allowing modules to be added, removed, or reordered without affecting the overall system. The pipeline is controlled by a configuration dictionary that specifies which modules are active, the substitution strength, the target TTS engine, and output file parameters.

Chapter 5: Dialect-Aware Pronunciation and Experimental Design

5.1 Mapping Phonological Rules to SSML Phoneme Annotations

Implementing dialect-aware pronunciation in a TTS system requires translating phonological rules into a format that speech synthesis engines can interpret. The present system achieves this through SSML phoneme annotations, following the approach validated by Gupta for Hindi dialects. The mapping between Khadi Boli phonological rules and SSML annotations is structured as a rule table that the system consults during SSML generation.

Gemination mapping. Consonant gemination is encoded using the IPA length marker /:/ appended to the affected consonant within the `<phoneme>` tag. For example, the Khadi Boli word बब्बा (babbā) receives the annotation `<phoneme alphabet="ipa" ph="bəb:ɑ:">बब्बा</phoneme>`, where the /b:/ indicates the lengthened geminate consonant. The SSML generator identifies candidate gemination sites by matching the Standard Hindi input against the gemination pattern (CVCV forms with specific consonants).

Retroflexion mapping. The retroflex lateral /ɭ/ and retroflex nasal /ɳ/ are specified using their IPA characters in phoneme annotations. For instance, काळ receives `<phoneme alphabet="ipa" ph="kɑ:ɭɑ:">काळ</phoneme>`. This annotation is critical because most TTS engines, trained on Standard Hindi, default to dental /l/ and /n/ realizations.

Vowel quality mapping. Vowel raising and dissipation patterns are encoded through modified IPA vowel symbols. The reduced vowel in Khadi Boli forms is annotated with the IPA breve marker for extra-short vowels where applicable.

The SSML generator is designed as a modular component that reads phonological rules from a JSON configuration file. Each rule specifies the target phoneme, the replacement phoneme, the phonological context (word-initial, word-medial, word-final, intervocalic), and the IPA representation. This design allows new rules to be added as the phonological analysis is refined through further fieldwork.

5.2 Dictionary-Driven Word Substitution

The dictionary-driven word substitution module forms the core of the dialect adaptation system. The module loads the 142-entry Khadi Boli dictionary into a Python dictionary data structure during initialization, mapping each Standard Hindi headword to its Khadi Boli equivalent. The substitution process operates at the token level: each token in the normalized input is looked up in the dictionary, and if a match is found, the token is replaced by its Khadi Boli equivalent.

The dictionary supports the following matching modes:

Exact match. The input token must exactly match the Standard Hindi form stored in the dictionary. This mode provides the highest precision but may miss inflected forms.

Stem match. The dictionary stores root forms, and the module applies Hindi morphological analysis to match inflected tokens against their dictionary root. This mode increases recall for verbs and nouns in various inflectional contexts.

Fuzzy match. For tokens not found in exact or stem matching, the module applies Levenshtein distance-based fuzzy matching with a configurable threshold. This mode handles spelling variations that arise from the absence of standardized Khadi Boli orthography.

The substitution module also maintains a frequency table derived from the corpus analysis (Section 2.4), which allows the system to prioritize substitutions for high-frequency dialect markers such as ते (replacing से) and कू (replacing को).

5.3 Unigram Frequency Comparison

The experimental design includes a quantitative comparison of lexical frequency patterns between Khadi Boli and Standard Hindi, implemented as a Jupyter notebook analysis. The comparison uses parallel corpora collected from both varieties: a Khadi Boli corpus of transcribed speech samples and a Standard Hindi corpus of comparable size and domain coverage. Both corpora are tokenized using the NLTK tokenizer with Devanagari character support.

The frequency analysis proceeds through the following steps. First, both corpora are cleaned by removing punctuation and normalizing whitespace. Second, token frequency distributions are computed for each corpus using NLTK's FreqDist class. Third, a merged vocabulary is constructed containing all tokens that appear in either

corpus, and aligned frequency vectors are created for the shared vocabulary. Fourth, the frequency differences are visualized using grouped bar charts and scatter plots. Finally, a linear regression model is fitted to predict Standard Hindi frequency from Khadi Boli frequency for the shared vocabulary.

This methodology is adapted from the approach of Gupta , who used linear regression to model the relationship between dialect scores across Hindi varieties. In the present work, the regression analysis serves to quantify the degree of lexical divergence between Khadi Boli and Standard Hindi, providing a baseline measure that can be tracked as the dictionary and corpus resources are expanded.

5.4 Linear Regression Analysis

The linear regression model fitted to the unigram frequency data provides a quantitative summary of the relationship between the two varieties. The model takes the form:

$$f_{SH} = \beta_0 + \beta_1 f_{KB} + \epsilon$$

where f_{SH} is the Standard Hindi unigram frequency, f_{KB} is the Khadi Boli unigram frequency, β_0 is the intercept, β_1 is the slope, and ϵ is the error term.

The fitted model yields an intercept $\beta_0 \approx 4.56$ and a slope $\beta_1 \approx 0.076$. The near-zero slope indicates that Khadi Boli frequency is a weak predictor of Standard Hindi frequency for the shared vocabulary. The positive intercept indicates that for words with minimal occurrence in Khadi Boli ($f_{KB} = 0$ or $f_{KB} = 1$), the Standard Hindi

frequency is expected to be substantially higher -- approximately 4 to 5 occurrences. This asymmetry reflects the broader lexical base of Standard Hindi, which maintains a larger inventory of low-frequency specialized vocabulary compared to the more compact lexical system of the spoken dialect.

Key diagnostics from the regression include the R^2 value, which indicates the proportion of variance in Standard Hindi frequency explained by Khadi Boli frequency. The low slope combined with the wide spread of residuals confirms that the two varieties, despite their shared origins and substantial lexical overlap, follow distinct frequency distributions that reflect their different functional and social roles.

5.5 Evaluation Methodology

While a full-scale human evaluation is beyond the scope of the present work, the experimental design includes a framework for future evaluation inspired by the methodologies of Gupta and Anand et al. . The proposed evaluation framework combines objective metrics with subjective assessment.

Objective metrics. The system performance is measured in terms of (1) word substitution accuracy, defined as the proportion of Khadi Boli dictionary entries correctly applied in test sentences; (2) SSML phoneme annotation coverage, defined as the proportion of phonological rule applications successfully encoded in SSML; and (3) pipeline processing time per sentence for each TTS engine.

Subjective evaluation (MOS/DMOS). The proposed subjective evaluation follows the Mean Opinion Score (MOS) and Degraded Mean Opinion Score (DMOS)

methodology used by Gupta . Native Khadi Boli speakers would rate synthesized speech samples on a five-point scale across dimensions including naturalness, intelligibility, dialect authenticity, and overall quality. DMOS comparison would contrast Khadi Boli-adapted output against Standard Hindi output for the same content, measuring the improvement attributable to the dialect adaptation modules.

Dialect authenticity scale. A novel evaluation dimension introduced in this work is dialect authenticity, which measures how closely the synthesized speech matches native Khadi Boli pronunciation. This dimension is particularly relevant because a TTS system can produce intelligible speech without capturing the distinctive phonological features of the target dialect. The dialect authenticity score is computed as the average of listener ratings on a five-point scale for the question: "Does this speech sound like a native Khadi Boli speaker?"

Chapter 6: Results and Discussion

6.1 Unigram Frequency Patterns

The unigram frequency analysis reveals several notable patterns in the lexical relationship between Khadi Boli and Standard Hindi. First, the shared vocabulary is dominated by high-frequency function words: the most frequent tokens in both corpora include के, है, हो, and का. These shared function words account for approximately 40% of all tokens in both varieties, confirming the genetic closeness of the two varieties at the level of grammatical vocabulary.

Second, the frequency distributions of content words show systematic divergence. Dialect-specific lexical items in Khadi Boli -- such as the postpositions ते and कू, the kinship terms बब्बा and ददा, and common verbs like खाणा and पैण्णी -- occupy frequency bands that are distinct from their Standard Hindi counterparts. In many cases, a Khadi Boli word shows higher frequency than its Standard Hindi equivalent, even though the Standard Hindi form is more widely used in written discourse. This pattern is consistent with the spoken nature of the Khadi Boli corpus: dialect-specific forms are favored in everyday speech, while written Standard Hindi tends toward a more formal register.

Third, the bar chart comparison (Figure 1) reveals that words with the largest frequency disparities are precisely those that function as dialect markers in everyday speech. The postposition के appears 9 times in the Khadi Boli corpus versus 1 time in the Standard Hindi corpus, while है shows the inverse pattern (5 vs. 9). These

asymmetries are not random but reflect systematic differences in grammatical marking between the spoken dialect and the written standard.

6.2 Regression Findings

The linear regression analysis (Figure 3) provides statistical confirmation of the qualitative observations discussed above. The fitted model shows three important characteristics:

Low slope ($\beta_1 \approx 0.076$). The near-zero slope indicates near-independence between the two frequency distributions. This finding has practical implications for TTS development: it confirms that a TTS system trained exclusively on Standard Hindi will not generalize well to Khadi Boli lexical patterns, and that dialect-specific resources are necessary for adequate performance.

Positive intercept ($\beta_0 \approx 4.56$). The positive intercept reflects the broader lexical diversity of Standard Hindi, which maintains a more extensive vocabulary with words distributed across a wider frequency range. Khadi Boli, as a primarily spoken variety, concentrates its lexical usage on a smaller core vocabulary, with many words appearing only once or twice in the corpus.

Wide residual variance. The large spread of residuals around the regression line indicates that even after accounting for the linear relationship, substantial lexical variation remains unexplained. This residual variation is the quantitative signature of dialect-specific lexical preferences that cannot be predicted from frequency alone --

precisely the kind of variation that the dictionary-driven substitution module is designed to handle.

These three findings converge on a clear conclusion: Khadi Boli and Standard Hindi, despite their close historical and genetic relationship, exhibit lexical frequency patterns that are sufficiently distinct to justify dedicated computational resources. The regression analysis provides a quantitative baseline against which future improvements in dialect coverage can be measured.

6.3 Dialect Variation Patterns

Beyond the frequency analysis, the phonological and morphological data reveal several broader patterns of dialect variation that inform the system design. First, the variation is systematic rather than sporadic: phonological rules such as gemination, retroflexion, and vowel dissipation apply consistently across the lexical inventory, and native speakers can reliably produce the dialect forms for novel words by applying these rules. This systematicity is what makes a rule-based approach to dialect-aware TTS feasible.

Second, the variation operates at multiple linguistic levels simultaneously. Phonological rules affect individual sounds, morphological rules affect word structure, and lexical selection affects whole vocabulary items. An effective dialect-aware TTS system must address all three levels. The present system addresses the lexical level through dictionary substitution, the phonological level through SSML phoneme annotations, and -- to a limited extent -- the morphological level through postposition replacement rules.

Third, the variation follows patterns that are consistent with the broader typology of Hindi dialect variation documented by Gupta and Rathore . Gemination is characteristic of both Khadi Boli and Haryanvi, while retroflexion shows different distributions across the western dialect group. The present findings for Khadi Boli align with the western group patterns identified by Rathore , who notes that Khadi Boli is classified within the western group of Hindi dialects alongside Haryanvi and Brajbhasha.

6.4 Challenges and Limitations

The development of the Khadi Boli TTS system encountered several significant challenges that are important to document for future work in low-resource dialect TTS.

Absence of a written corpus. Khadi Boli is primarily a spoken dialect with no standardized orthographic tradition. The lack of a written corpus meant that all lexical data had to be collected through fieldwork, transcribed manually, and verified with native speakers. The 142-entry dictionary, while valuable as a first resource, represents only a fraction of the full Khadi Boli lexicon. The corpus used for frequency analysis was limited in size, and the frequency patterns should be interpreted as suggestive rather than definitive.

Multiple pronunciations per word. Many Khadi Boli words exhibit variation in pronunciation across speakers and even within the speech of a single speaker. For example, the word for 'water' was documented as both पैणी and पाणी in different recordings. This variation presents a challenge for a TTS system that must select a

single pronunciation for each word in the input text. The system handles this by storing alternate forms in the dictionary and selecting among them based on context when possible.

Extra stress patterns. The characteristic extra stress on the middle of Khadi Boli words is difficult to replicate with existing TTS engines, which control prosody at the level of whole utterances rather than individual syllables. While SSML prosody tags provide some control, the fine-grained stress patterns of the dialect exceed what current SSML standards can express. This limitation is inherent to the use of general-purpose TTS engines rather than a custom-trained model.

Engine limitations for dialect-specific phonemes. Neither gTTS nor pyttsx3 accept SSML phoneme annotations, limiting the dialect adaptation for these engines to word substitution alone. Only the `edge_tts` engine with the Madhur or Swara voices provides full SSML support. This means that the quality of dialect adaptation varies significantly across the three engines, with `edge_tts` offering the most authentic Khadi Boli output and gTTS offering the least.

6.5 Implications for Low-Resource Dialect TTS

Despite these challenges, the present work demonstrates that a functional dialect-aware TTS system can be constructed for a low-resource dialect using existing TTS infrastructure, provided that sufficient linguistic analysis and lexical resources are developed. The key enabling factors are (1) the systematic nature of phonological and lexical variation, which allows rule-based adaptation to complement dictionary-driven substitution; (2) the availability of cloud-based TTS engines with SSML

support, which provide high-quality base synthesis that can be adapted through phonetic annotations; and (3) the modular pipeline architecture, which allows each component to be improved independently as resources grow.

The approach developed here is portable to other low-resource dialects of Hindi and potentially to dialects of other major languages with similar TTS infrastructure. The general methodology -- linguistic documentation, dictionary construction, phonological rule formalization, SSML-based pronunciation adaptation, and quantitative corpus analysis -- can be applied to any dialect for which a minimum of lexical and phonological data is available.

Chapter 7: Conclusion and Future Work

7.1 Summary of Contributions

This paper has presented the first text-to-speech system specifically designed for Khadi Boli, a low-resource dialect of Hindi spoken in western Uttar Pradesh. The work makes four major contributions to the field of low-resource dialect speech technology.

First computational resource for Khadi Boli. The dictionary of over 142 Khadi Boli words and phrases, compiled through fieldwork in the Budhana and Muzaffarnagar region, represents the first systematic written documentation of Khadi Boli vocabulary. Each entry is documented with its Khadi Boli form, Standard Hindi equivalent, English translation, and Kruti Dev encoding representation. The dictionary covers a broad range of semantic domains including common nouns, verbs, kinship terms, time expressions, and directional markers.

Comprehensive linguistic analysis. The phonological analysis documented in Chapter 2 identifies six systematic phonological rules that distinguish Khadi Boli from Standard Hindi: consonant gemination, retroflexion, postposition replacement, initial consonant replacement, alpa-pran-maha-pran consonant interaction, and vowel dissipation (Balaghat tendency). The morphological analysis identifies distinctive patterns in verbal agreement, postpositional marking, and pronominal systems. These linguistic insights provide the foundation for the dialect adaptation modules in the TTS system.

Working TTS implementation. The implemented TTS system integrates three synthesis engines (gTTS, edge_tts with Madhur/Swara voices, and pyttsx3) with a modular pipeline architecture that includes text normalization, dialect-aware word substitution, SSML parsing with Devanagari support, and sentence tokenization. The system is fully functional and produces synthesized speech in Khadi Boli with varying degrees of dialect adaptation depending on the engine selected.

Quantitative corpus analysis. The unigram frequency comparison and linear regression analysis provide quantitative evidence of systematic lexical divergence between Khadi Boli and Standard Hindi. The regression model (intercept $\beta_0 \approx 4.56$, slope $\beta_1 \approx 0.076$) confirms that the two varieties exhibit substantially different lexical frequency patterns, validating the need for dedicated dialect resources.

7.2 Limitations

The present work has several limitations that should be candidly acknowledged. The Khadi Boli dictionary, while valuable as an initial resource, is limited to 142 entries and covers only a fraction of the full lexical inventory of the dialect. A comprehensive dictionary would require substantially more fieldwork and would likely number in the thousands of entries. The corpus used for frequency analysis is similarly limited in size and coverage; the unigram frequency patterns reported here should be considered indicative rather than definitive.

The TTS system's dialect adaptation capabilities are constrained by the capabilities of the underlying TTS engines. Full SSML-based phonetic adaptation is only available through the edge_tts engine. The gTTS and pyttsx3 engines, while valuable for their

accessibility and offline capabilities respectively, do not support the fine-grained pronunciation control needed for authentic Khadi Boli output. The system currently lacks a formal human evaluation (MOS/DMOS study), which would be necessary to quantify the perceptual improvement attributable to the dialect adaptation modules.

Finally, the system has not been tested across the full range of Khadi Boli variation. The dialect varies across geographic areas within the Budhana and Muzaffarnagar region, and the phonological rules documented here may not apply uniformly across all Khadi Boli-speaking communities.

7.3 Future Directions

Several directions for future work emerge from this project.

Expanded lexical resources. The most urgent need is for a larger, more comprehensive dictionary of Khadi Boli. A target of 1,000-2,000 entries would provide substantially better coverage and enable more robust dialect adaptation. The dictionary could be expanded through systematic fieldwork, community collaboration, and analysis of recorded speech samples. The expansion would also enable improved corpus construction for more reliable frequency analysis.

Neural TTS training. The ultimate solution for authentic Khadi Boli TTS would be to train a neural TTS model on Khadi Boli speech data. This would require collecting a speech corpus of at least several hours from native Khadi Boli speakers, along with transcribed text. With such a corpus, a Tacotron- or FastSpeech-based model could be trained to produce speech that captures the full phonological and prosodic

characteristics of the dialect, including the extra stress patterns that are difficult to replicate with SSML-based approaches.

Expanded dialect coverage. The methodology developed in this work can be extended to other Hindi dialects, particularly those in the western group (Haryanvi, Brajbhasha) and eastern group (Awadhi, Bihari) identified by Rathore . A multi-dialect TTS platform could enable dialect-aware speech synthesis across the entire Hindi dialect continuum, using a common architecture with dialect-specific dictionaries and phonological rule sets.

Formal evaluation study. A formal MOS/DMOS evaluation with native Khadi Boli speakers would provide quantitative evidence of the system's effectiveness. Following the methodology of Gupta and Anand et al. , the evaluation would compare dialect-adapted output against Standard Hindi output, measuring naturalness, intelligibility, and dialect authenticity. Such an evaluation would guide further refinements to the dialect adaptation modules.

Integration with ASR and machine translation. The Khadi Boli resources developed in this work -- the dictionary, phonological rules, and corpora -- could also support automatic speech recognition (ASR) and machine translation for the dialect. An ASR system trained with Khadi Boli pronunciation rules would better recognize dialect speech, and a translation system could support Khadi Boli to Standard Hindi conversion. These applications would extend the benefits of language technology to Khadi Boli speakers in domains beyond speech synthesis.

In conclusion, this work demonstrates that a meaningful step toward dialect-aware speech technology for low-resource varieties is achievable through careful linguistic analysis, strategic use of existing TTS infrastructure, and targeted computational implementation. The Khadi Boli TTS system, while imperfect, represents a foundation upon which increasingly capable dialect speech technology can be built.

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